

URL Shortener service – Complete Testing Guide

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1. Pre-requisites

Requirement	Version	How to Check
Python	3.11 or higher	python --version
pip	Any (comes with python)	pip --version

Note:

- Windows: Download Python from <https://www.python.org/> -> During install, check “Add to Path”
- Mac: Open Terminal -> brew install python3 or download from <https://www.python.org/>

Command Differences:

OS	Python Command	Pip command
Windows	python	pip
Mac	Python3	Pip3

```
hp@LAPTOP-3JLRI300 MINGW64 ~  
• $ python --version  
Python 3.13.14  
  
hp@LAPTOP-3JLRI300 MINGW64 ~  
• $ pip --version  
pip 26.1.2 from C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.13_3.13.3824.0_x64__qbz5n2kfra8p0\Lib\site-packages  
\pip (python 3.13)  
.
```

2. Setup

Step 1: Download the Project folder and Open Terminal (Command Prompt or VS Code) and go to Project folder and install Dependencies, start the server, open the app in browser.

cd url-shortener-service

pip install -r requirements.txt

```
hp@LAPTOP-3JLRI300 MINGW64 ~/Downloads/hemanth/url-shortener-service  
• $ ll  
total 15  
drwxr-xr-x 1 hp 197121  0 Jul 30 05:50 app/  
drwxr-xr-x 1 hp 197121  0 Jul 30 05:50 docs/  
-rw-r--r-- 1 hp 197121 550 Jul 30 05:50 main.py  
-rw-r--r-- 1 hp 197121 8052 Jul 30 05:50 README.md  
-rw-r--r-- 1 hp 197121  38 Jul 30 05:50 requirements.txt  
-rw-r--r-- 1 hp 197121 162 Jul 30 05:50 START_HERE.txt  
drwxr-xr-x 1 hp 197121  0 Jul 30 05:50 tests/  
  
hp@LAPTOP-3JLRI300 MINGW64 ~/Downloads/hemanth/url-shortener-service  
• $ pip install -r requirements.txt
```

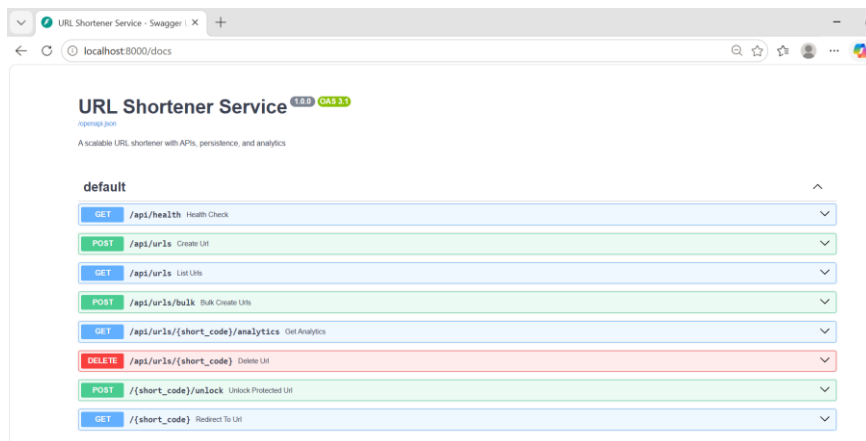
Start the server:

python main.py

```
hp@LAPTOP-3JLRI300 MINGW64 ~/Downloads/hemanth/url-shortener-service
○ $ python main.py
INFO: Will watch for changes in these directories: ['C:\\Users\\hp\\Downloads\\hemanth\\url-shortener-service']
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Started reloader process [10356] using StatReload
INFO: Started server process [10460]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Open the App in browser:

Go to: <http://localhost:8000/docs>



Step-3: Feature Testing

Feature 1: Health Check

**** Business Case:**** verify the server is running

**** Steps:****

1. In UI, click `GET /api/health`
2. Click "Try it out"
3. Click "Execute"

**** Expected Output:**** "status": "healthy"

Request URL	
<code>http://localhost:8000/api/health</code>	
Server response	
Code	Details
200	Response body<pre>{ "status": "healthy"}</pre>

Feature 2: Create Short URL

**** Business Case:**** Converts a long website link into a short, easy-to-share link.

****Steps:****

1. Click `POST /api/urls`
2. Click "Try it out"
3. Clear the text box, paste:

```
{
  "url": "https://www.schwab.com",
  "custom_code": "schwab"
}
```

4. Click "Execute"

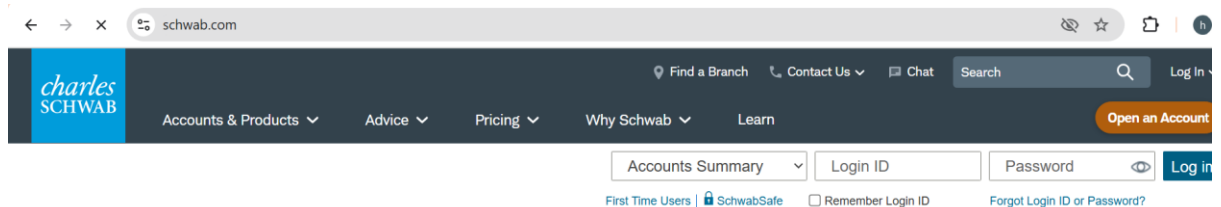
**** Expected Output:****

Server response	
Code	Details
201	Response body<pre>{ "short_code": "schwab", "short_url": "http://localhost:8000/schwab", "original_url": "https://www.schwab.com", "created_at": "2026-07-30 02:19:08", "max_clicks": null, "expires_at": null, "is_protected": false }</pre>

5. Test the short URL (Customer Clicks short link -> arrives at real website).

In browser, type: <http://localhost:8000/schwab>

Browser redirects to Schwab official Home Page.



Feature 3: List all URLs

**** Business Case:**** Admin Dashboard showing all active links.

****Steps:****

1. Click `POST /api/urls`
2. Click "Try it out"

Code	Details
200	Response body<pre>{ "urls": [{ "short_code": "schwab", "short_url": "http://localhost:8000/schwab", "original_url": "https://www.schwab.com", "created_at": "2026-07-30 02:19:08", "max_clicks": null, "expires_at": null, "is_protected": false }] }</pre>

Feature 4: Bulk Create URLs

****Business Case:**** Company might have 100+ product pages. Creating them one by one takes hours. Bulk create does it in one click. (Short url codes will be auto generated)

****steps:****

1. Click `POST /api/urls/bulk`
2. Click "Try it out"
3. Clear the text box, paste:

```
{
  "urls": [
    "https://www.schwab.com/trading",
    "https://www.schwab.com/retirement"
  ]
}
```

POST /api/urls/bulk Bulk Create Urls

Create multiple short URLs at once (max 50).

Parameters

No parameters

Request body required

Edit Value | Schema

```
{
  "urls": [
    "https://www.schwab.com/trading",
    "https://www.schwab.com/retirement"
  ]
}
```

4. Click "Execute"

Server response

Code	Details
201	Response body <pre>{ "created": 2, "results": [{ "short_code": "AJ5nHFq", "short_url": "http://localhost:8000/AJ5nHFq", "original_url": "https://www.schwab.com/trading", "created_at": "2026-07-30 03:11:52", "max_clicks": null, "expires_at": null, "is_protected": false }, { "short_code": "GSyPl3H", "short_url": "http://localhost:8000/GSyPl3H", "original_url": "https://www.schwab.com/retirement", "created_at": "2026-07-30 03:11:52", "max_clicks": null, "expires_at": null, "is_protected": false }] }</pre>

5. Multiple Short URLs will be created. We can test in browser with short URL. It will redirect to original Page.

<http://localhost:8000/AJ5nHFq> -> <https://www.schwab.com/trading>

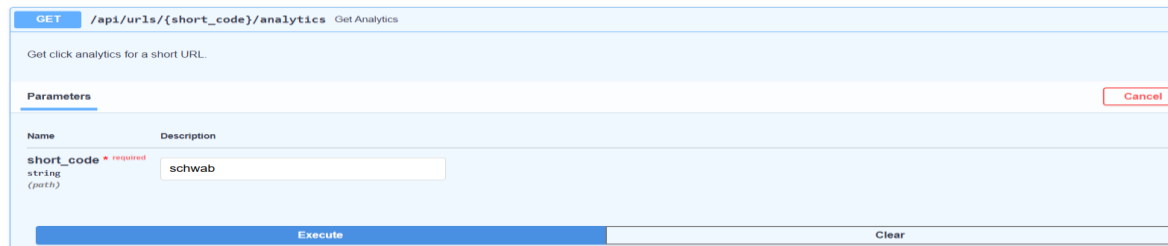
<http://localhost:8000/GSyPl3H> -> <https://www.schwab.com/retirement>

Feature 5: Analytics

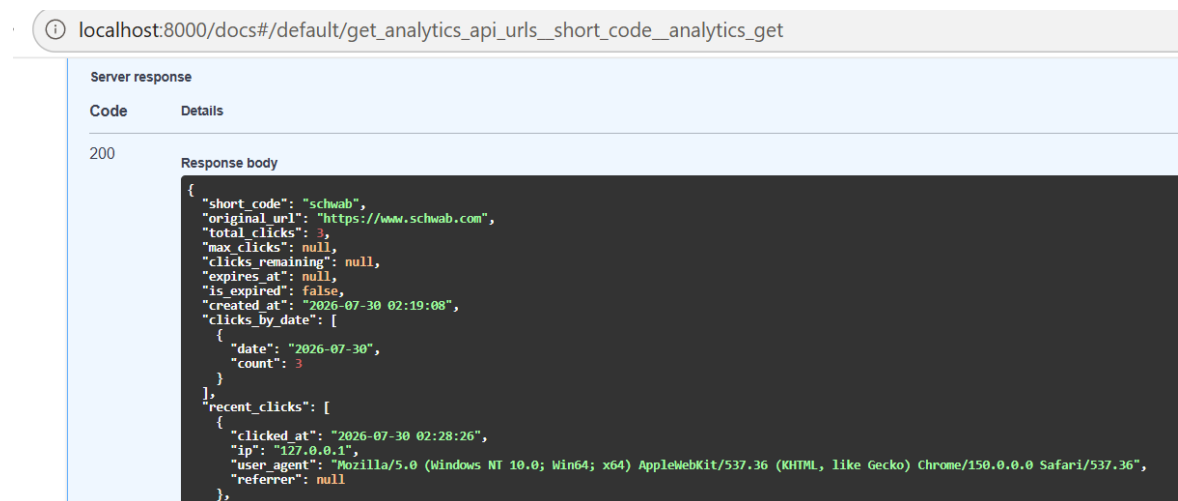
**** Business Case:**** It helps in understanding customer engagement – how many people are interested and interacting with the shared links.

****Steps:****

1. Click `POST /api/urls/{short\_code}/analytics`
2. Click "Try it out"
3. Type short url word "schwab" in the short_code field
4. Click "Execute"



The screenshot shows a REST client interface with a GET request to `/api/urls/{short_code}/analytics`. The description says "Get click analytics for a short URL". Under the "Parameters" tab, there is a table with two columns: "Name" and "Description". The first row has "short_code" as the name, marked as "required" and "string (path)". The value "schwab" is entered in the input field next to it. At the bottom, there are "Execute" and "Clear" buttons.



The screenshot shows the "Server response" section of the REST client. The status code is 200. The response body is a JSON object:

```
{
  "short_code": "schwab",
  "original_url": "https://www.schwab.com",
  "total_clicks": 3,
  "max_clicks": null,
  "clicks_remaining": null,
  "expires_at": null,
  "is_expired": false,
  "created_at": "2026-07-30 02:19:08",
  "clicks_by_date": [
    {
      "date": "2026-07-30",
      "count": 3
    }
  ],
  "recent_clicks": [
    {
      "clicked_at": "2026-07-30 02:28:26",
      "ip": "127.0.0.1",
      "user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36",
      "referrer": null
    }
  ]
}
```

We can see how many people clicked our product link, at what time they clicked and from which location. It helps in understanding customer engagement – how many people are interested and interacting with the shared links.

Feature 6: Delete URL

**** Business Case:**** Removes a link that no longer needed.

****Steps:****

1. Click `POST /api/urls/{short\_code}`
2. Click "Try it out"
3. Type short url word "schwab" in the short_code field
4. Click "Execute"

localhost:8000/docs#/default/delete_url_api_urls_short_code_delete

DELETE /api/urls/{short_code} Delete Url

Delete a short URL (soft delete).

Parameters

Name	Description
short_code * required string (path)	schwab

Execute

Expected: Code 204 (no body means short URL deleted)

Verify: Open <http://localhost:8000/schwab> in browser - should show "not found" now.

← ↻ ⓘ localhost:8000/schwab

Pretty-print ☐

```
{"detail": "Short URL not found, expired, or click limit reached"}
```

Feature 7: Unlock Protected URL – Password Protection

**** Business Case:**** This is necessary to share a confidential link like a salary document – only people with the password can access the confidential document.

****Steps:****

- Create a password protected URL:
Go to POST /api/urls -> Try it out -> Paste:


```
{
  "url": "https://www.schwab.com/wealth-management",
  "custom_code": "wealth-advisor",
  "password": "schwab123"
}
```

POST /api/urls Create Url

Create a new short URL. Options: max_clicks, expires_in_hours, password.

Parameters

No parameters

Request body required

Edit Value | Schema

```
{
  "url": "https://www.schwab.com/wealth-management",
  "custom_code": "wealth-advisor",
  "password": "schwab123"
}
```

2. Try – Unlocking:
Go to POST `/short_code/unlock` – Try it out: Enter the short code and password.
Short-code: wealth-advisor
Passwords: schwab123

Request URL	
<code>http://localhost:8000/wealth-advisor/unlock</code>	
Server response	
Code	Details
200	Response body <pre>{ "original_url": "https://www.schwab.com/wealth-management" }</pre>

Provides a secure way to access password-protected links – verifies the password and reveals the original URL if correct.

Feature 8: Click Limit

**** Business Case:**** A link that stops working after a set number of clicks – like “only first 3 people can access the offer.”

****Steps:****

1. Go to POST `/api/urls` -> Try it out -> Paste:


```
{
  "url": "https://www.schwab.com/",
  "custom_code": "schwab",
  "max_clicks": "3"
}
```
2. Click Execute – you get 201 (success)
3. To Test, Go to GET `/short_code` -> Try it out

Request URL	
<code>http://localhost:8000/schwab</code>	
Server response	
Code	Details
404 <i>Undocumented</i>	Error: Not Found Response body <pre>{ "detail": "Short URL not found, expired, or click limit reached" }</pre>

Feature 9: Rate Limiting (Anti-Spam protection)

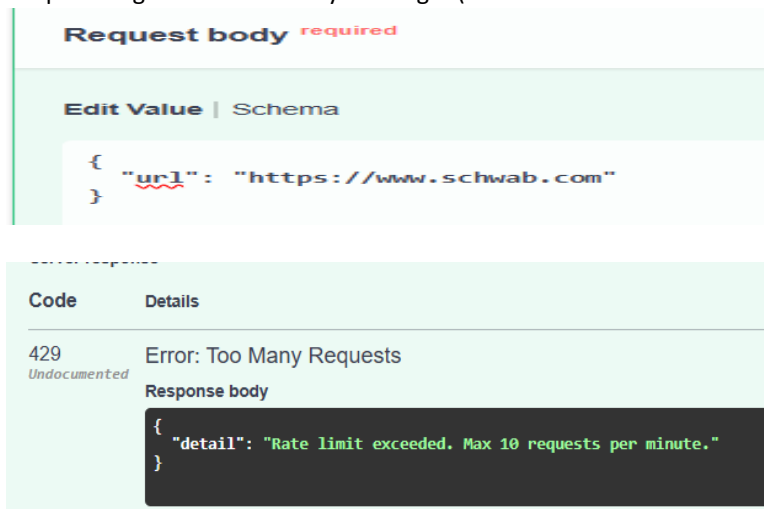
**** Business Case:**** If one person keeps creating too many links too fast, the system will block them for 1 minute. This keeps the system safe and fair for everyone.

We have configured In a way 10 requests per minute.

****Steps:****

4. Go to POST /api/urls -> Try it out -> Paste:

```
{  
  "url": "https://www.schwab.com/"  
}
```
5. Click Execute – you get 201 (success)
6. Keep clicking after 10 times – you will get (Rate limit exceeded. Max 10 requests per minute)



This can be used to Automatically blocks users who make too many requests – Protects the system from spam, bots.

Feature 10: URL Blocking (security)

Business case: Prevents our platform from being misused to spread harmful links – if someone tries to shorten a dangerous website, the system automatically rejects it.

****Steps:****

1. Go to POST /api/urls -> Try it out -> Paste:

```
{  
  "url": "https://malware.com/steal-data"  
}
```

POST **/api/urls** Create Url

Create a new short URL. Options: max_clicks, expires_in_hours, password.

Parameters

No parameters

Request body required

Edit Value | **Schema**

```
{  
  "url": "https://malware.com/steal-data"  
}
```

As we can see, the request is rejected by our system.

Server response	
Code	Details
409 <i>Undocumented</i>	Error: Conflict Response body <pre>{ "detail": "URL is blocked - domain is not allowed" }</pre> Response headers <pre>content-length: 51 content-type: application/json date: Thu, 30 Jul 2026 12:39:55 GMT server: uvicorn</pre>

Automatically checks every URL against a list of known dangerous websites and rejects them – keeps the platform safe from misuse.

4. Automated Tests:

To check if all features are working correctly

****Steps:****

1. Open a new terminal
2. Go to project folder
3. Run: **python -m pytest tests/ -v**

Expected: all test cases passed (all green)

```
hp@LAPTOP-3JLRI300 MINGW64 ~/Downloads/hemanth/url-shortener-service
$ python -m pytest tests/ -v
===== test session starts =====
platform win32 -- Python 3.13.14, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\hp\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_
qbz5n2kfra8p0\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\hp\Downloads\hemanth\url-shortener-service
plugins: anyio-4.14.2
collected 27 items

tests/test_api.py::test_create_short_url PASSED [ 3%]
tests/test_api.py::test_create_with_custom_code PASSED [ 7%]
tests/test_api.py::test_create_invalid_url PASSED [ 11%]
tests/test_api.py::test_create_duplicate_code PASSED [ 14%]
tests/test_api.py::test_redirect PASSED [ 18%]
tests/test_api.py::test_redirect_not_found PASSED [ 22%]
tests/test_api.py::test_analytics PASSED [ 25%]
tests/test_api.py::test_analytics_not_found PASSED [ 29%]
tests/test_api.py::test_list_urls PASSED [ 33%]
tests/test_api.py::test_delete_url PASSED [ 37%]
tests/test_api.py::test_delete_not_found PASSED [ 40%]
tests/test_api.py::test_health PASSED [ 44%]
tests/test_api.py::test_create_with_click_limit PASSED [ 48%]
tests/test_api.py::test_click_limit_works PASSED [ 51%]
tests/test_api.py::test_click_limit_shows_in_analytics PASSED [ 55%]
tests/test_api.py::test_no_click_limit_works_forever PASSED [ 59%]
tests/test_api.py::test_create_with_expiration PASSED [ 62%]
tests/test_api.py::test_non_expired_url_works PASSED [ 66%]
tests/test_api.py::test_create_with_password PASSED [ 70%]
tests/test_api.py::test_password_required_on_access PASSED [ 74%]
tests/test_api.py::test_wrong_password_rejected PASSED [ 77%]
tests/test_api.py::test_correct_password_works PASSED [ 81%]
tests/test_api.py::test_unlock_endpoint PASSED [ 85%]
tests/test_api.py::test_bulk_create PASSED [ 88%]
tests/test_api.py::test_bulk_create_with_expiry PASSED [ 92%]
tests/test_api.py::test_blocked_domain_rejected PASSED [ 96%]
tests/test_api.py::test_safe_domain_allowed PASSED [100%]
```

5. Troubleshooting:

Problem	Solution
`python: command not found`	Try `python3` instead`
500 Internal server error	Delete `data/` folder and restart server
422 Error	Check JSON – no trailing commas after last value
Port 8000 in use	Close other apps using port 8000, or restart